

define $\vec{p} = m\vec{v}$

Rearr Newton's Law $\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt}$

Therefore $F = \frac{d}{dt}(m\vec{v}) = \frac{d\vec{p}}{dt}$

Actually, this is the deeper version, since

$$\frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) = \frac{dm}{dt}\vec{v} + m \frac{d\vec{v}}{dt}$$

Why do $\sum \vec{F}_{\text{internal}} = 0$?

what we focused on
before assumes $\dot{m} = 0$

→ "If I pull my hair hard"

I can't lift myself up and levitate.

Idea: Center of Mass

$$\vec{R} = \frac{1}{M} \sum m_i \vec{r}_i \quad \text{for discrete system}$$

$$\vec{R} = \frac{1}{M} \int \vec{r} dm \quad \text{for continuous system}$$

More than half the stars in Milky Way
are made of Binary stars.

Outside Charleston,

there is an apple tree,
who claims to be direct descendant of Newton's
apple tree.

Earth-Moon



"The moon doesn't
orbit the Earth
It orbits the COM."

$$10^7 \text{ metres} = \frac{1}{\pi} (2\pi R_E)$$



$$R_E = \frac{2}{\pi} \times 10^7 \text{ m.}$$

$$= \frac{20}{\pi} \text{ Mega-meters.}$$

$$\frac{R_{\text{com}}}{R_{\text{Earth}}} \sim 0.6$$

$$F = \frac{G M m}{r^2 \dots}$$

why is it 2?

why not $2\pi = 2.00000 \dots 1$?

"If you could measure that, you would get
a Nobel prize."

What is the Period of Earth's Mean motion?

$$F = \frac{GMm}{r^2}$$

$$F = \frac{mv^2}{r} = mr\omega^2$$

$$\frac{GMm}{r^2} = mr \left(\frac{2\pi}{T} \right)^2$$

recall

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$\Rightarrow \left(\frac{T}{2\pi} \right)^2 = \frac{r^3}{GM}$$

$$T = 2\pi \sqrt{\frac{r^3}{GM}} \sim 29 \text{ days}$$

"How many seconds in year?"
 $\pi \cdot 10^7$

Jan wrote

$$\vec{R} = \lim_{N \rightarrow \infty} \frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i$$

$$= \frac{1}{M} \int \vec{r} dm$$

$$= \frac{1}{M} \int \vec{r} \rho dV$$

$$= \frac{1}{M} \int dx \int dy \int dz \rho(\vec{r}) \vec{r}$$

$$= \frac{1}{M} \int d^3\vec{x} \vec{x} \rho(\vec{x})$$



take a body
chop it into smaller
pieces

This limit becomes an Integral

$$\rho = \frac{dm}{dV}$$

$$\int dV = \int dx \int dy \int dz = \int d^3\vec{x}$$

in physics, it doesn't matter if we
write

$$\int [\dots] dx \quad \text{or} \quad \int dx [\dots]$$

they are the same